

## **Model Selection and Preprocessing Write-Up**

This project uses a binary classification setup to predict whether an individual earns more than \$50K per year. The positive class is >50K and the negative class is ≤50K, and the business objective is to identify high-income prospects while limiting wasted outreach. Because the dataset is imbalanced, with a positive class rate of 23.93% and a negative class rate of 76.07%, accuracy alone is not a reliable measure of success. For this reason, F1 score was selected as the primary metric because it balances precision and recall, which directly reflects the business trade-off between catching true high-income cases and avoiding incorrect outreach. The majority baseline confirms this issue: although it reaches 76.07% accuracy by predicting everyone as ≤50K, it produces 0.00 precision, 0.00 recall, and 0.00 F1. The education rule baseline performs better, with an F1 score of 0.491, but it still misses a substantial number of true positives. These results establish that the task is not about maximizing raw accuracy; it is about finding the best balance between identifying genuine high-income individuals and controlling false positives.

For preprocessing, I used a standard and defensible strategy that separates numeric and categorical features. Numeric columns were handled with median imputation followed by StandardScaler, while categorical columns were handled with most-frequent imputation and OneHotEncoder(handle\_unknown='ignore'). This choice was appropriate because the dataset contains a mix of continuous variables and highly skewed numeric features, where median imputation is more robust than mean imputation in the presence of outliers. In addition, scaling is important because some models are sensitive to feature magnitude, and standardization ensures that numeric variables are on a comparable scale. For categorical variables, OneHotEncoder is preferable to ordinal encoding because the categories are nominal rather than ordered. This preserves category meaning without imposing an artificial ranking. Alternative methods such as mean imputation, label encoding, dropping incomplete rows, or applying target encoding were considered but skipped because they would either be more sensitive to skew, distort category relationships, or introduce leakage or interpretability problems.

The model comparison showed that the preprocessing pipeline works well for both a linear model and a tree-based model, but the two approaches are useful for different reasons. Logistic regression is the primary model I will continue developing because it is highly interpretable and provides direct insight into which features are increasing or decreasing the probability of the positive class after one-hot encoding. In practice, the high-weight coefficients correspond to meaningful variables such as education, marital status, and work-related indicators, which makes the model explainable to stakeholders. This is especially important in a business-facing problem where understanding why a person is classified as high-income matters as much as raw performance. The decision tree is still useful as a nonlinear benchmark because it can capture interactions that a linear model may miss, but it is more prone to overfitting. The decision tree consistently showed a larger train-test gap, and its top splits reflected more specific combinations that were less stable and less obviously aligned with domain knowledge. This suggests that while the tree can be informative, it may not generalize as reliably as the logistic model.

For Day 3, I will continue with the logistic regression pipeline as the main model, while retaining the decision tree as a comparison model and a check on nonlinearity. I will also keep the preprocessing pipeline reusable so that future experiments begin from a stable and consistent foundation. The next

preprocessing improvements I plan to test are targeted rather than broad: log transformations for heavily skewed variables such as capital-gain and capital-loss, grouping of rare categories in occupation and native-country, and more careful handling of missing categorical values. These steps should help reduce noise and improve generalization without disrupting the underlying structure of the model. Overall, the pipeline-based workflow has already improved substantially over the majority and rule baselines, and the logistic regression model is the best choice for continued development because it offers the strongest balance of interpretability, stability, and business relevance.